1.1 What Are Cells?

Biology · Cell Structure and Organisation · Syllabus refs (G2 & G3): 1a, 1b, 1c

Name: _____ Class: _____ Date: _____

Learning outcomes

- Identify and state the functions of the cell structures (including organelles) of typical plant and animal cells from diagrams, light micrographs and as seen under the light microscope using prepared slides and fresh material treated with an appropriate temporary staining technique: cell wall, cell membrane, cytoplasm, nucleus, cell vacuoles (large, sap-filled in plant cells; small, temporary in animal cells), chloroplasts.
- Identify and state the functions of the following organelles from diagrams and electron micrographs: mitochondria, ribosomes.
- Compare the structure of typical animal and plant cells.

Trigger activity

Watch this real video taken through a microscope. A *Paramecium* is a living thing made of just **one cell**. As you watch, think: **how can one tiny cell move, feed and stay alive on its own?** What parts must it have?

What is a cell?

Every living thing is made of **cells**, just as a building is made of bricks. Cells are the **building blocks of life**: the simplest structural and functional units of life. Some organisms, such as bacteria, are a single cell. You are made of about 30 trillion cells.

Inside a cell there are smaller structures with special jobs, called **organelles** (“little organs”). The nucleus, mitochondria and chloroplasts are organelles.

All the living material of a cell is called **protoplasm**. It is made up of three key parts:

cell membrane + cytoplasm + nucleus

> **Key idea:** structure is linked to function. Each part of a cell has a shape and make-up that suits its job.

In a large organism, cells are organised in levels. Similar cells work together as a **tissue**, tissues form an **organ**, organs work together in an **organ system**, and the systems make up the whole **organism**.

cell → tissue → organ → organ system → organism

For example: a muscle cell → muscle tissue → the heart → the circulatory system → you.

Seeing cells: light and electron microscopes

Cells are too small to see with the naked eye, so we use microscopes. A photo taken through a microscope is called a **micrograph**.



Light microscope: the kind in your school lab. Light passes through a thin specimen.



Electron microscope: a large machine in a research lab. A beam of electrons passes through the specimen.

Photos: GcG (public domain, numbers removed); Gannu03, CC BY-SA 4.0, via Wikimedia Commons.

	Light microscope	Electron microscope
Uses	light	a beam of electrons
Picture	light micrograph	electron micrograph
Where you meet it	the school lab	research labs (you will see its pictures)
Structures you can see	cell wall, cell membrane, cytoplasm, nucleus, vacuole, chloroplasts	all of these **plus** the small organelles: **mitochondria** and **ribosomes**

> **Remember:** mitochondria and ribosomes are too small to see clearly with a school light microscope. You identify them from diagrams and electron micrographs.

Practical: making a temporary slide of onion cells

1. Peel a thin, see-through layer (the epidermis) from the inside of a piece of onion.
2. Lay it flat in a drop of water on a clean glass slide. Use a mounting needle so it does not fold.
3. Add a drop of **iodine solution**. The stain makes the nucleus and cell wall easier to see.
4. Lower a cover slip **slowly, at an angle**, using the mounting needle. This stops air bubbles being trapped.
5. Soak up extra liquid at the edge with filter paper. Look at the slide under **low power** first, then high power.

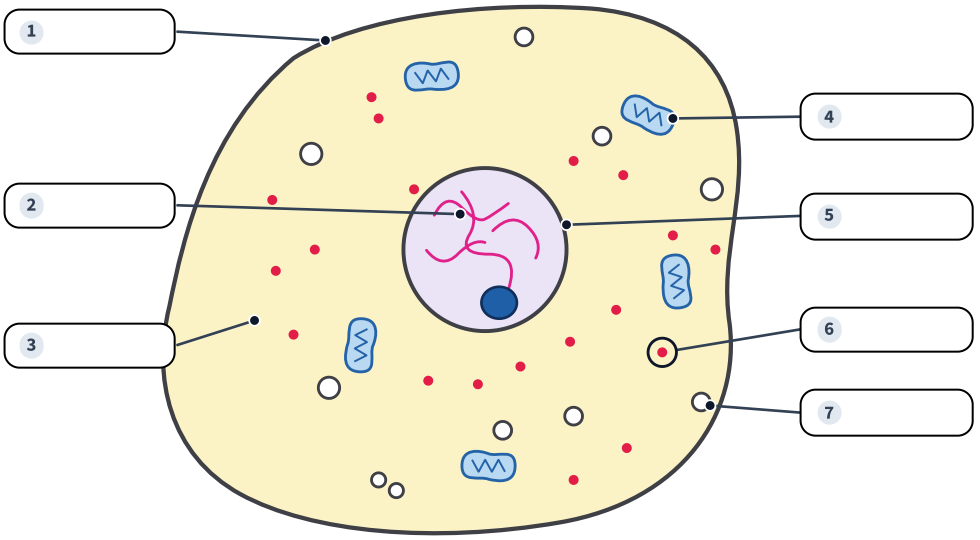
Animal cells (for example cheek cells) are stained with **methylene blue**. Follow your school's safety rules for this practical.

Label the cells

Drag each label to the right box. Then press **Check my labels**. Wrong labels turn red: move them and check again.

Label a typical animal cell.

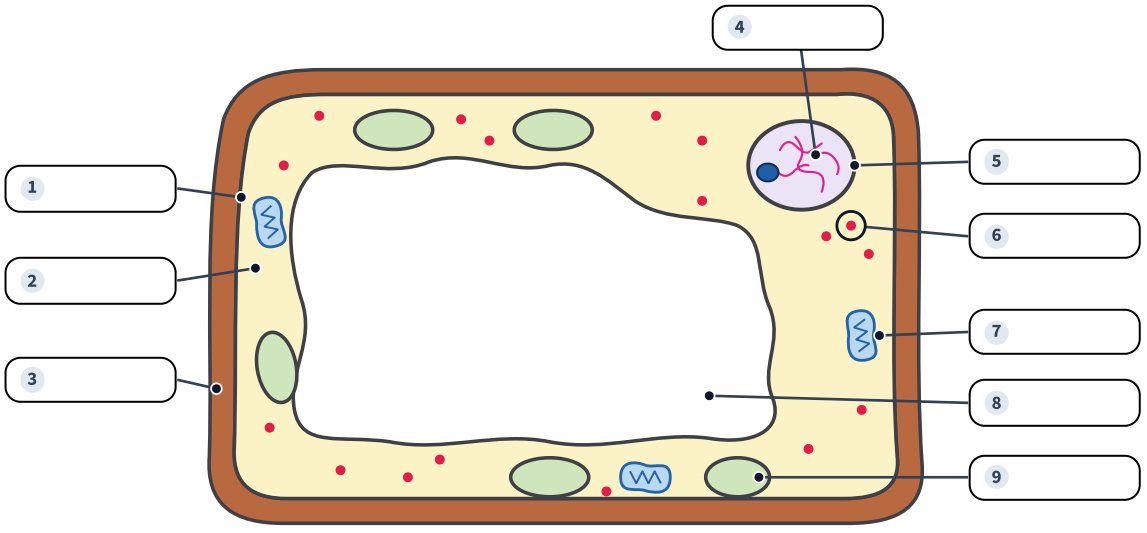
cell wall | nucleus | ribosome | mitochondrion | chloroplast | cell membrane | vacuole | chromosome | cytoplasm



A typical animal cell (not to scale).

Label a typical plant cell.

cytoplasm | chloroplast | chromosome | cell membrane | mitochondrion | vacuole | ribosome | cell wall | nucleus



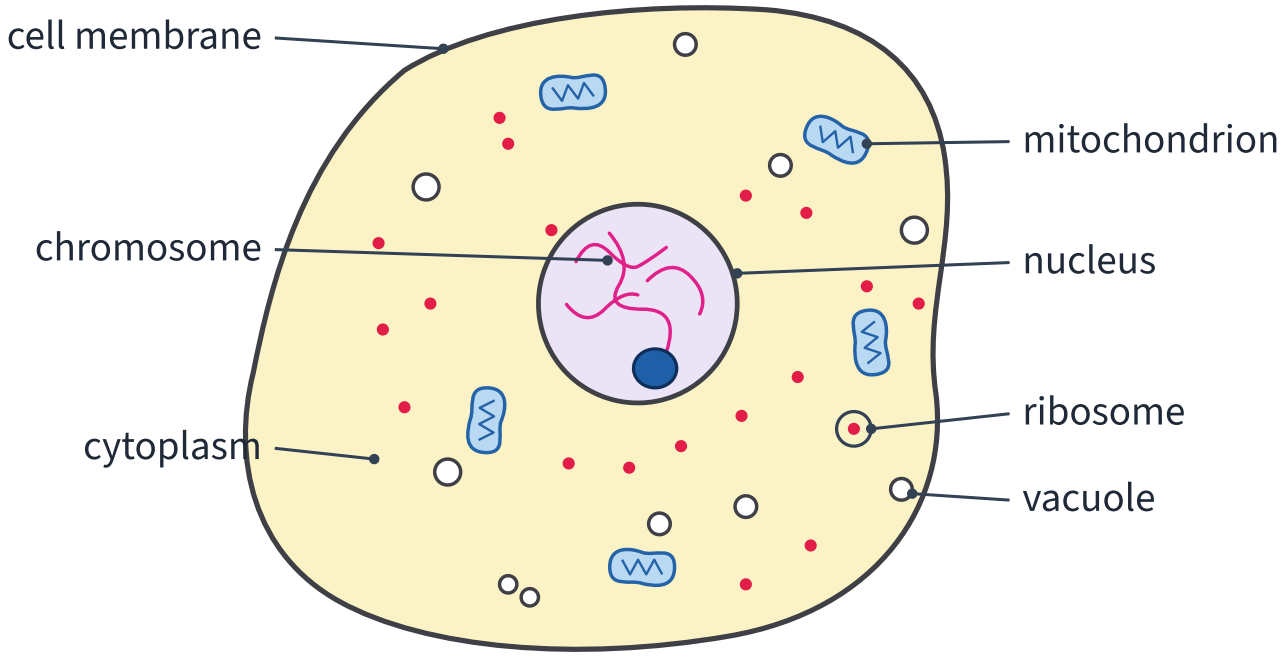
A typical plant cell, e.g. a leaf cell (not to scale).

Cell structures and their functions

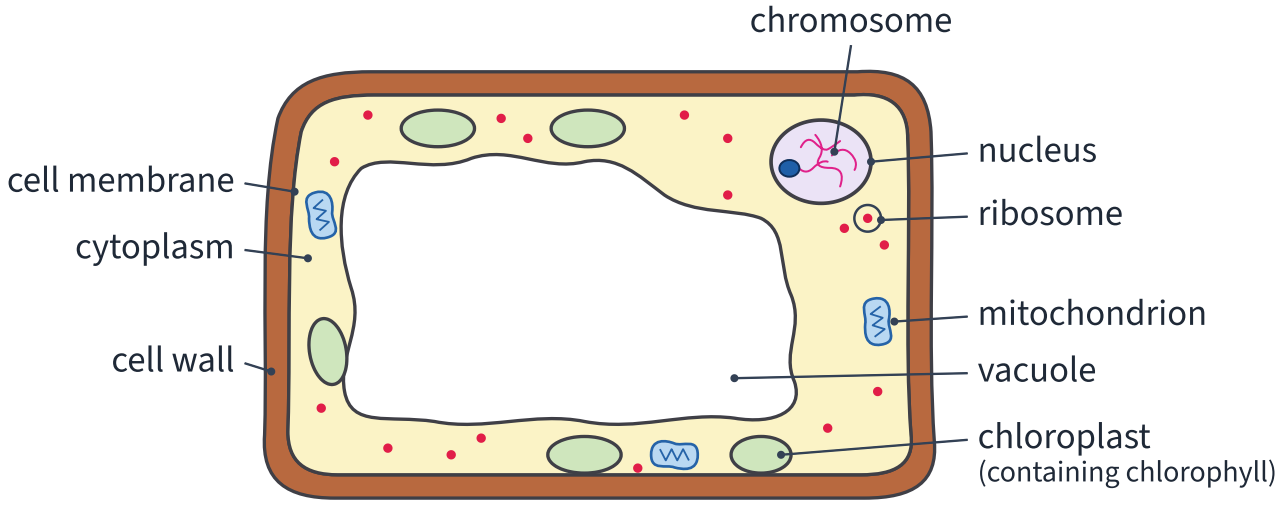
Structure	Function	Found in
Cell membrane	Controls the movement of substances into and out of the cell. It is **partially permeable**: only small, soluble substances can pass through.	plant and animal cells
Cell wall	Protects the cell from injury and gives it a **fixed shape**. Made of **cellulose**. It is **fully permeable**.	plant cells only
Cytoplasm	The site of cellular activities. Contains the organelles.	plant and animal cells
Nucleus	Controls cell activities such as cell division, growth and repair. Contains **chromosomes** made of **DNA** (deoxyribonucleic acid), which carries hereditary information.	plant and animal cells
Vacuole	Stores substances. Plant cell: one large central vacuole of **cell sap** (sugar, mineral salts, amino acids). Animal cell: small, temporary vacuoles of water and food.	plant and animal cells
Mitochondria	The site of **aerobic respiration**, which releases energy for the cell.	plant and animal cells
Ribosomes	The site of **protein synthesis**.	plant and animal cells
Chloroplasts	Contain the green pigment **chlorophyll**, which absorbs light for **photosynthesis** (how plants make food).	some plant cells

Common mistakes: the cell *wall* is fully permeable; the cell *membrane* is partially permeable. Mitochondria *release* energy by respiration; they do not "make" energy. Animal cells never have a cell wall.

Comparing plant and animal cells



Animal cell



Plant cell

What they have in common

Both plant and animal cells have a **cell membrane**, **cytoplasm**, a **nucleus**, **mitochondria**, **ribosomes** and **vacuoles**. These structures do the same jobs in both kinds of cell.

How they are different

Plant cell	Animal cell
Cell wall **present** (made of cellulose)	Cell wall **absent**
Chloroplasts **present** (in many plant cells)	Chloroplasts **absent**
One **large, central** vacuole containing cell sap	**Numerous small**, temporary vacuoles
Fixed, regular shape (because of the cell wall)	No fixed shape

> **Careful:** not every plant cell has chloroplasts. Root cells and onion bulb cells grow in the dark, so they have none. A cell with a cell wall but no chloroplasts can still be a plant cell. The cell wall is the surest sign.

How to answer a "compare" question

In an exam, **compare** means give **similarities and differences**. Write each difference as a matched pair, so both cells are mentioned.

Weak answer

"A plant cell has a cell wall and chloroplasts."

Only one cell is described, and there is no similarity.

Strong answer

"Both cells have a nucleus, cytoplasm and a cell membrane. A plant cell has a cellulose cell wall, *but* an animal cell does not. A plant cell has one large central vacuole, *whereas* an animal cell has many small vacuoles."

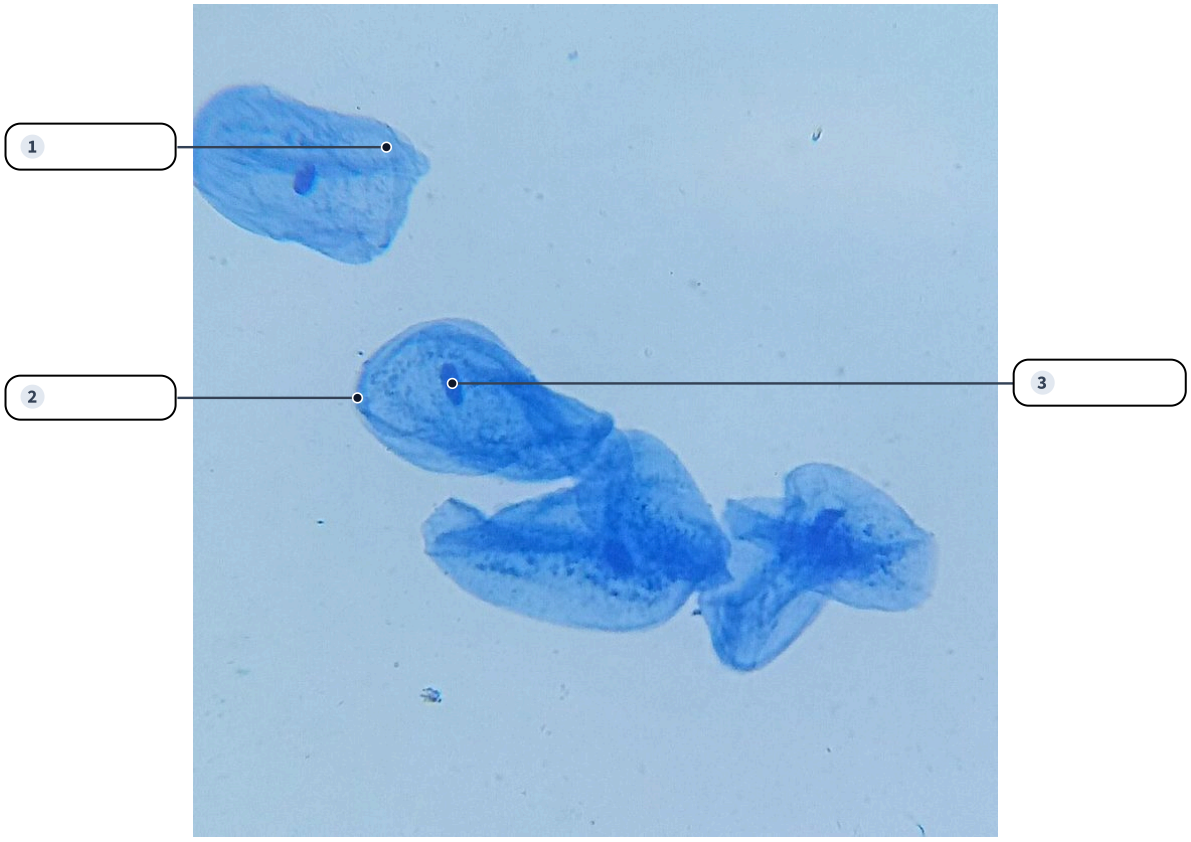
Label real cells: light micrographs

These are real photos taken through a light microscope. Real cells look less tidy than diagrams. Use what you know: a **cell wall** gives plant cells a fixed, box-like shape; a stained **nucleus** shows

up as a dark spot; **chloroplasts** are green.

Human cheek cells (animal cells)

cell membrane | nucleus | cytoplasm

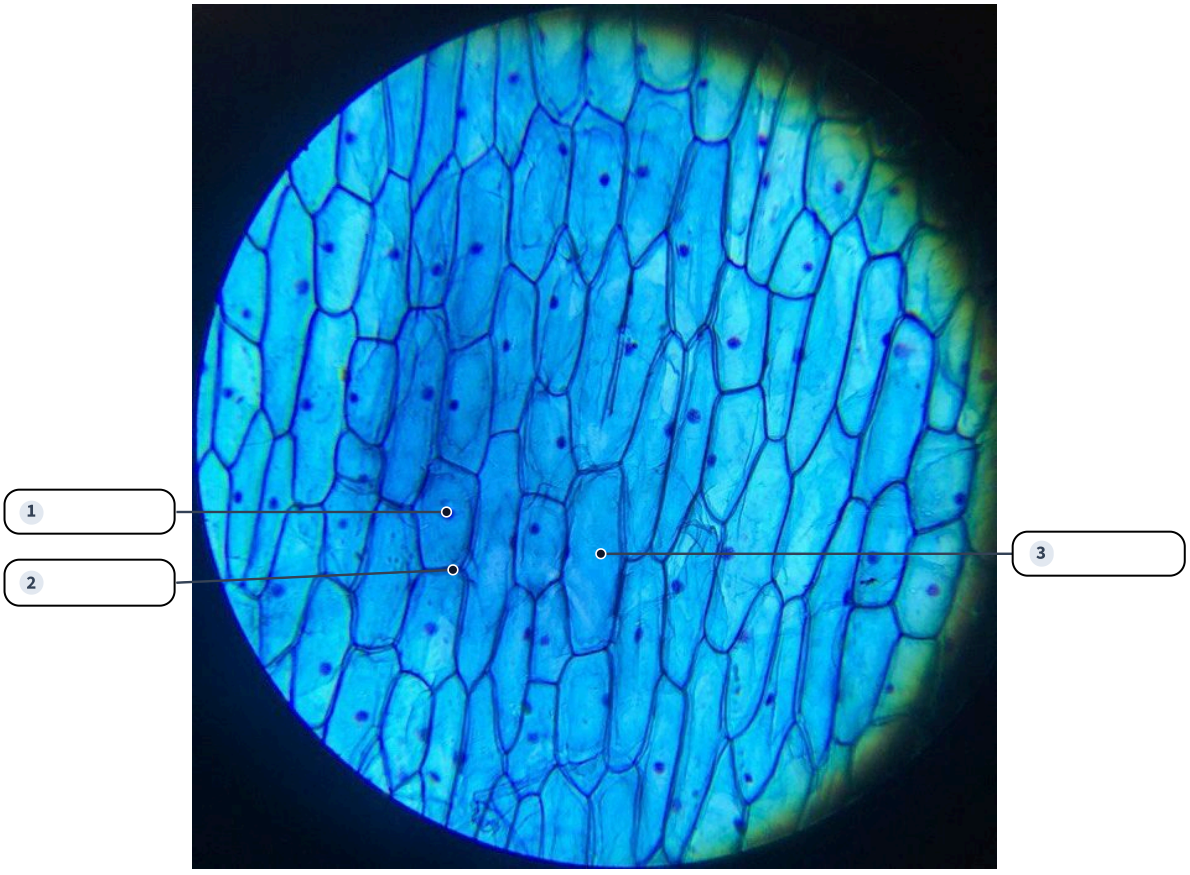


Human cheek cells stained with methylene blue, light microscope.

Photo: Fritzmann2002, CC BY-SA 4.0, via Wikimedia Commons.

Onion epidermis cells (plant cells)

cell wall | vacuole | nucleus

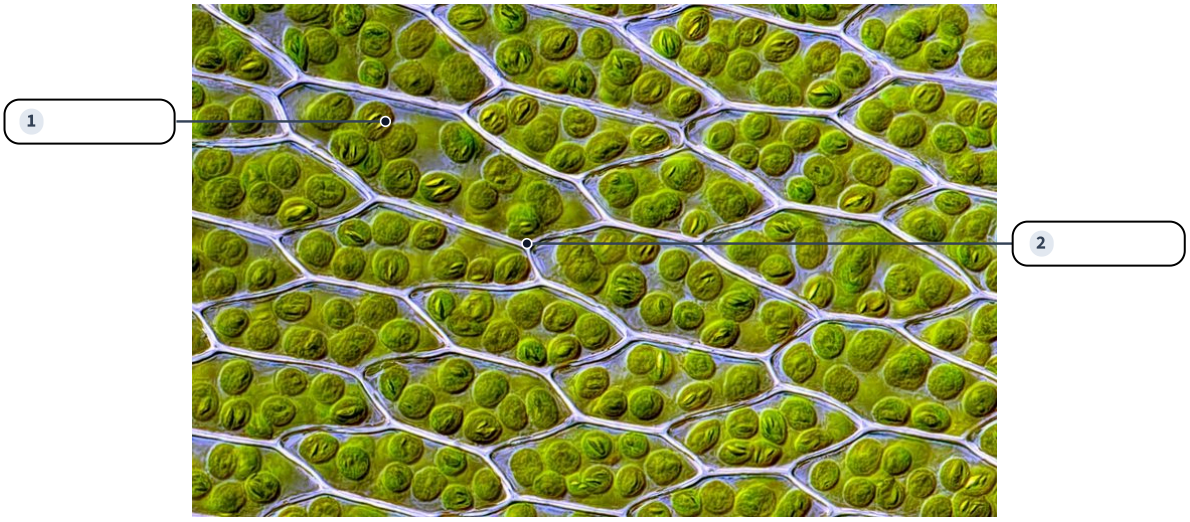


Onion epidermal cells, stained, light microscope. Notice: no chloroplasts.

Photo: Katekor1, CC BY 4.0, via Wikimedia Commons.

Moss leaf cells (plant cells)

chloroplast | cell wall



Leaf cells of a moss, light microscope (no stain needed).

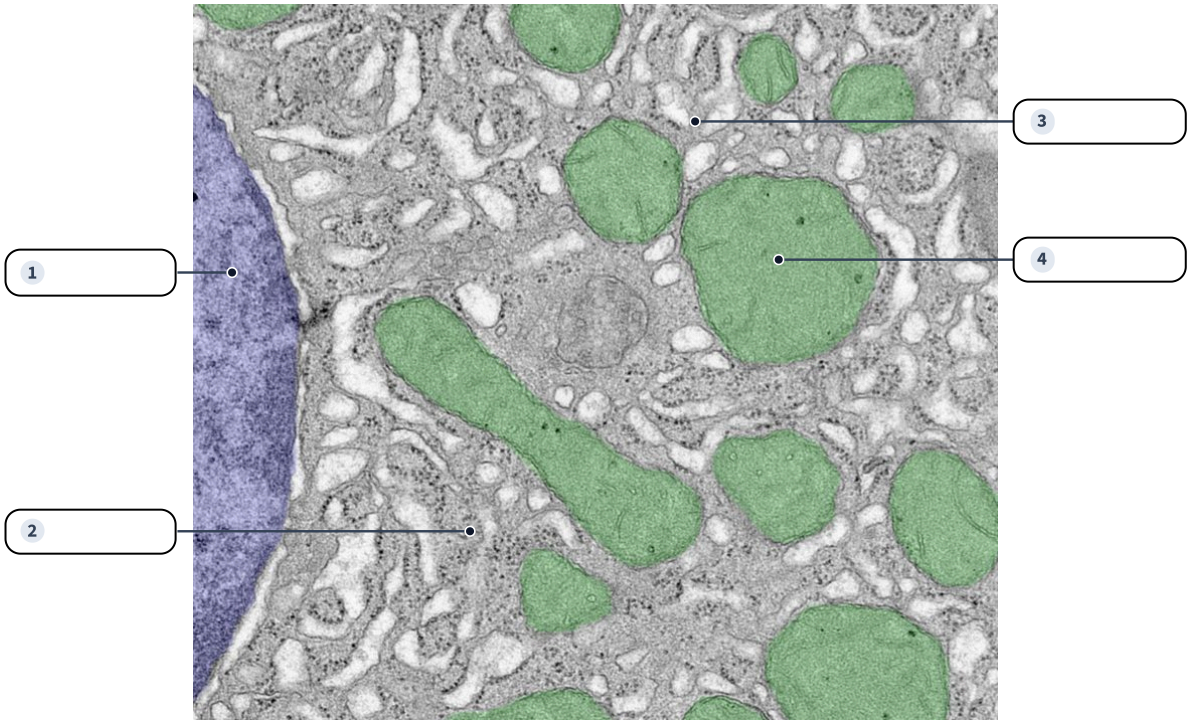
Photo: Des_Callaghan, CC BY-SA 4.0, via Wikimedia Commons.

Label real cells: electron micrographs

An electron microscope shows much more detail, so you can see the small organelles. Electron micrographs are black and white; colour is sometimes added afterwards to make parts easier to see.

Part of a liver cell (animal cell)

mitochondrion | ribosomes | nucleus | cytoplasm



Part of a liver cell, electron micrograph. The nucleus (purple) and mitochondria (green) have been coloured. The tiny dark dots are ribosomes.

Image (cropped): Bochimoto H, Matsuno N, Ishihara Y et al., CC BY 4.0, via Wikimedia Commons.

Part of a leaf cell (plant cell)

chloroplast | cell wall | vacuole



Part of a leaf cell, electron micrograph (about 12 000 times bigger than life). The lines inside the chloroplast are where light is absorbed.

Image: and3k and caper437, CC BY-SA 3.0, via Wikimedia Commons.

Spotting mitochondria and chloroplasts

In the exam you may be shown an electron micrograph and asked to name the organelle. These two are the ones to know well. They look quite different once you know what to look for.

Mitochondria

One whole mitochondrion cut across.

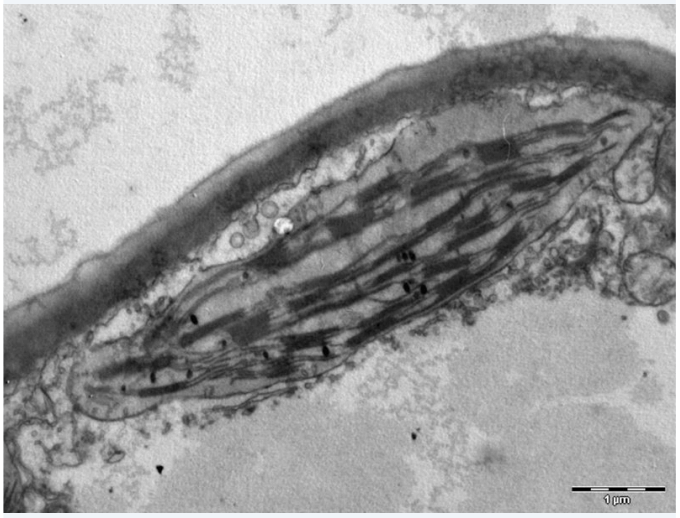
One whole mitochondrion cut along its length.

- **Shape:** small ovals or sausages (often round when cut across).
- **Edge:** a smooth double membrane.
- **Inside:** many folds of the inner membrane, like shelves or fingers. The folds give a large surface for aerobic respiration.
- **Where:** in the cytoplasm of almost every plant and animal cell. Very active cells (muscle, liver) have lots.

Images: Louisa Howard, Dartmouth Electron Microscope Facility (public domain, via Wikimedia Commons); Prof. R. Bellairs, CC BY 4.0, Wellcome Collection.

Chloroplasts

A whole chloroplast in a bean leaf cell (colour added).



A whole chloroplast in a leaf cell, lying against the cell wall.

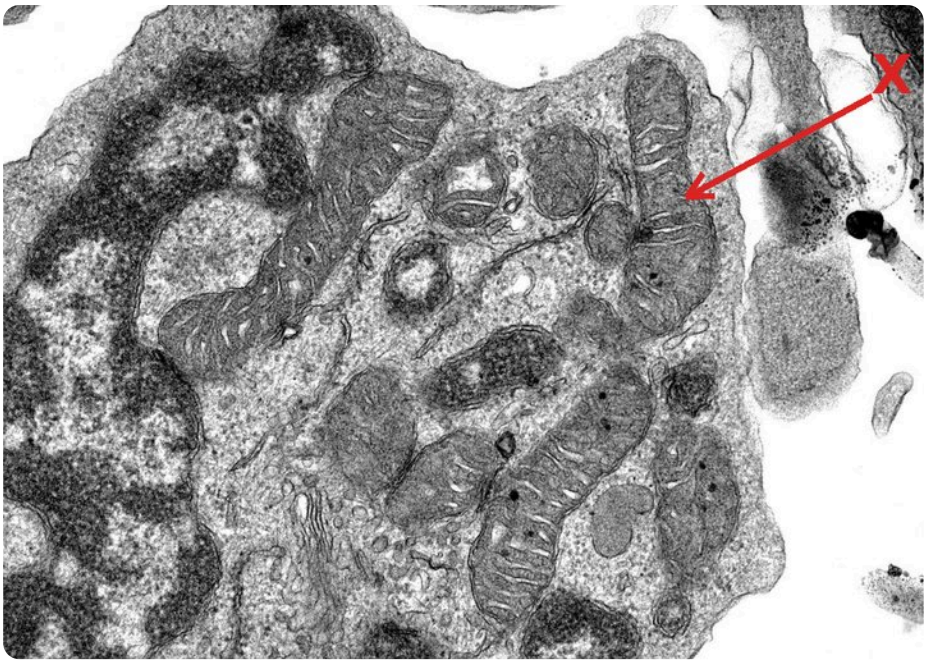
- **Shape:** a large lens or oval, much bigger than a mitochondrion.
- **Inside:** long, parallel membranes with dark **stacks** (like piles of coins). Chlorophyll is in these membranes, where light is absorbed.
- **Where:** only in some plant cells (e.g. leaf cells), usually near the cell wall.

Images: Kevin Mackenzie, University of Aberdeen, CC BY 4.0, Wellcome Collection (cropped); and3k and caper437, CC BY-SA 3.0, via Wikimedia Commons.

Look for…	Mitochondrion	Chloroplast
Size	small	large (several times longer)
Membranes inside	folds that stick in from the edge	long parallel membranes with dark stacks
Found in	plant and animal cells	some plant cells only
Job	aerobic respiration	photosynthesis

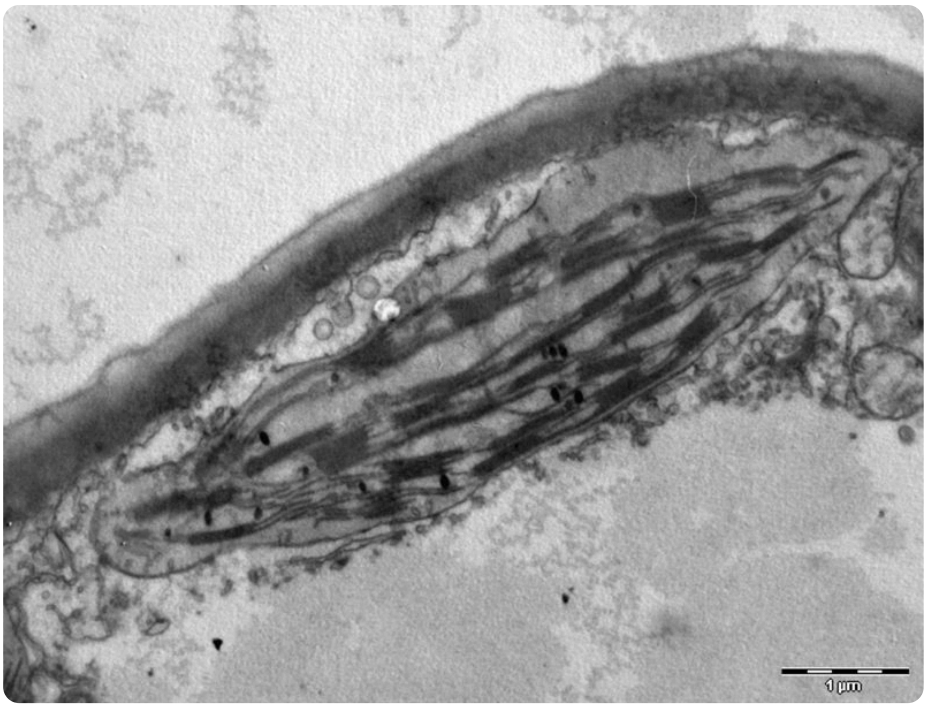
Spot the organelle

A. This electron micrograph shows part of an animal cell. Name the organelle labelled X. (There are many more of them in this cell.)



- ○ Nucleus
- ○ Mitochondrion
- ○ Chloroplast
- ○ Vacuole

B. Name the organelle in this electron micrograph.



- ○ Mitochondrion
- ○ Vacuole
- ○ Nucleus
- ○ Chloroplast

C. A student finds an organelle with stacks of membranes in an electron micrograph of a cell. What can the student conclude?

- ○ The cell is an animal cell.
- ○ The cell is a plant cell, such as a leaf cell.
- ○ The cell has no nucleus.
- ○ The cell is a red blood cell.

Question A image: Dr Jeremy Skepper, CC BY 4.0, Wellcome Collection (arrow added). Question B image: credited above.

Check your understanding

1. Which structure controls the movement of substances into and out of a cell?

- ○ Cell wall
- ○ Cell membrane
- ○ Cytoplasm
- ○ Nucleus

2. Which structure contains chromosomes made of DNA?

- ○ Nucleus
- ○ Ribosome
- ○ Vacuole
- ○ Chloroplast

3. Where is protein made in a cell?

- ○ Mitochondria
- ○ Cell sap
- ○ Ribosomes
- ○ Nucleus

4. A student looks at a cheek cell under a school light microscope. Which structure will the student NOT be able to see clearly?

- ○ Nucleus
- ○ Cell membrane
- ○ Cytoplasm
- ○ Mitochondria

5. Why is the cover slip lowered slowly, at an angle, when making a slide?

- ○ To stain the cells
- ○ To stop air bubbles being trapped
- ○ To make the cells bigger
- ○ To kill the cells

6. Which three structures are found in plant cells but NOT in animal cells?

- ○ Cell wall, chloroplasts, large central vacuole
- ○ Cell membrane, nucleus, cytoplasm
- ○ Mitochondria, ribosomes, cell wall
- ○ Chloroplasts, nucleus, ribosomes

7. A cell has a cell wall and a large vacuole but no chloroplasts. Where could it come from?

- ○ A human cheek
- ○ A plant root
- ○ A human muscle
- ○ A red blood cell

8. Which statement about animal cells is correct?

- ○ They have one large central vacuole.
- ○ They have small, temporary vacuoles.
- ○ They have a cell wall made of cellulose.
- ○ They have chloroplasts.

9. Why do both plant and animal cells have mitochondria?

- ○ Both need to photosynthesise.
- ○ Both need to release energy by aerobic respiration.
- ○ Both need a fixed shape.
- ○ Both store cell sap.

Exam practice: try the structured question comparing a **liver cell** and a **root hair cell** (Section B, question 6) on the Revision Questions page.